Lecture #13: The Bivariate and Multivariate Normal Distributions

MATH 321 – Mathematical Statistics I

Torin Quinlivan

Knox College
tquinlivan@knox.edu

November 9, 2026

KNOX COLLEGE

Normal Distribution

We've now dealt with the single-variable Normal distribution a lot.

$$X \sim \text{Normal}(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

How might we generalize this to several variables?

KNOX COLLEGE

The Bivariate Normal Distribution

We could use independence.

Suppose that:

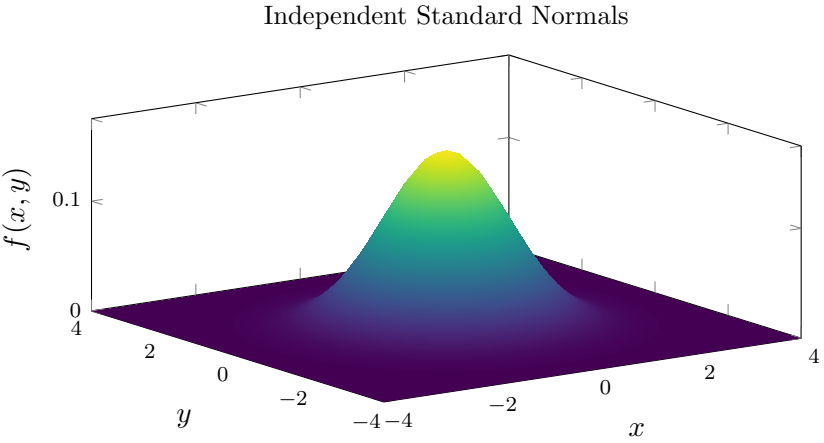
$$X \sim \text{Normal}(\mu_1, \sigma_1^2)$$
$$Y \sim \text{Normal}(\mu_2, \sigma_2^2)$$

If X and Y are independent, then:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left\{ -\frac{1}{2} \left[\left(\frac{x-\mu_1}{\sigma_1} \right)^2 + \left(\frac{y-\mu_2}{\sigma_2} \right)^2 \right] \right\}$$

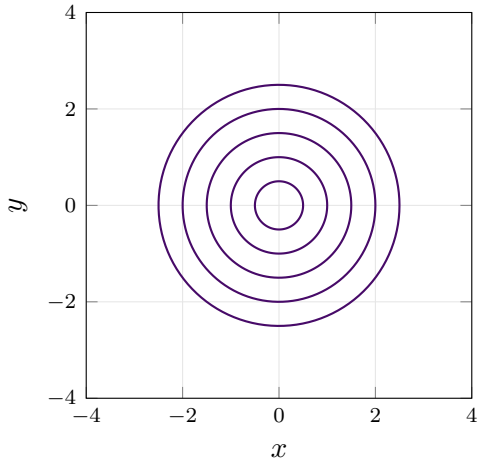
KNOX COLLEGE

The Bivariate Normal Distribution



The Bivariate Normal Distribution

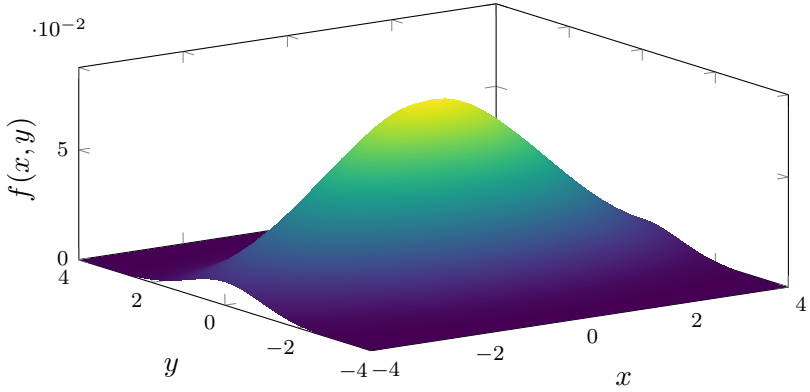
Contours of $f(x, y)$: Independent Standard Normals



KNOX COLLEGE

The Bivariate Normal Distribution

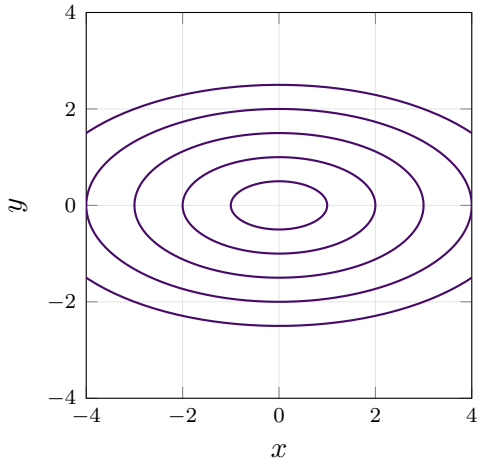
Independent: $X \sim \text{Normal}(0, 4)$, $Y \sim \text{Normal}(0, 1)$



KNOX COLLEGE

The Bivariate Normal Distribution

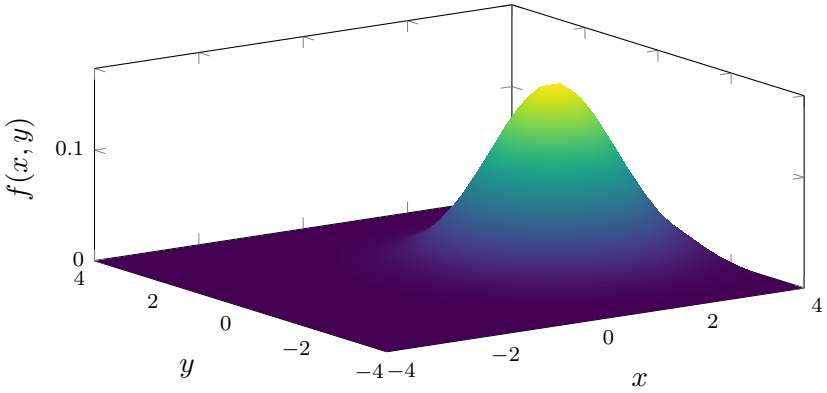
Contours: $X \sim \text{Normal}(0, 4)$, $Y \sim \text{Normal}(0, 1)$



KNOX COLLEGE

The Bivariate Normal Distribution

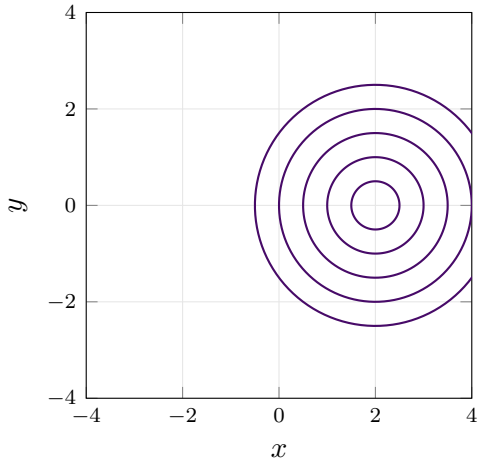
Independent: $X \sim \text{Normal}(2, 1)$, $Y \sim \text{Normal}(0, 1)$



KNOX COLLEGE

The Bivariate Normal Distribution

Contours: $X \sim \text{Normal}(2, 1)$, $Y \sim \text{Normal}(0, 1)$

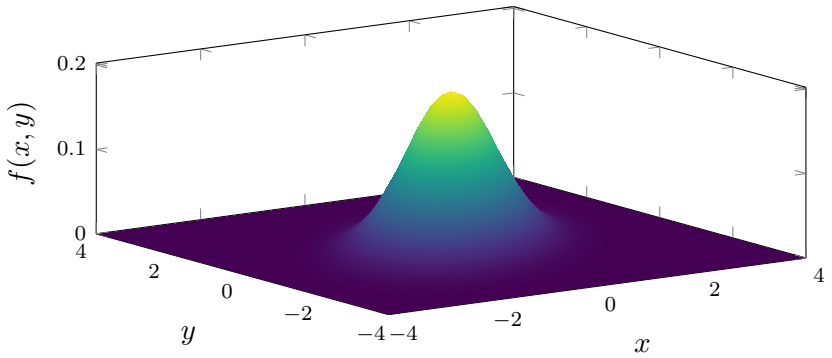


KNOX COLLEGE

The Bivariate Normal Distribution

But what if X and Y are not independent?

Bivariate Normal with $\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$

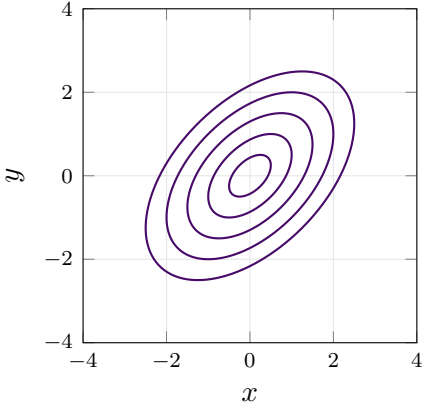


KNOX COLLEGE

The Bivariate Normal Distribution

But what if X and Y are not independent?

Contours with $\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$



KNOX COLLEGE

The Bivariate Normal Distribution

Two random variables X and Y are said to be **bivariate normal**, or **jointly normal**, if $aX + bY$ has a Normal distribution for all $a, b \in \mathbb{R}$. (Here a constant counts as a Normal random variable with variance 0.)

KNOX COLLEGE

The Standard Bivariate Normal Distribution

Two random variables X and Y are said to have the **standard bivariate normal distribution with correlation coefficient** ρ if their joint PDF is given by

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left(x^2 - 2\rho xy + y^2\right)\right\},$$

where $\rho \in (-1,1)$. Each of X and Y then has mean 0 and variance 1, and ρ is their correlation. If $\rho = 0$, then we just say X and Y have the **standard bivariate normal distribution**.

KNOX COLLEGE

The Bivariate Normal Distribution

Two random variables X and Y are said to have the **bivariate normal distribution with parameters μ_X, σ_X^2, μ_Y, σ_Y^2 and ρ** if their joint PDF is given by

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left\{ -\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right)\right]\right\}$$

where $\mu_X, \mu_Y \in \mathbb{R}$, $\sigma_X, \sigma_Y > 0$ and $\rho \in (-1,1)$ are all constants.

KNOX COLLEGE

The Bivariate Normal Distribution

Theorem: Let X and Y be bivariate normal random variables with parameters μ_X, σ_X^2, μ_Y, σ_Y^2 and ρ. Then there exist independent standard normal random variables Z_1 and Z_2 such that

$$X = \sigma_X Z_1 + \mu_X$$
$$Y = \sigma_Y \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_Y.$$

KNOX COLLEGE

Conditional Expectation and Variance

Theorem: Suppose X and Y are jointly normal random variables with parameters μ_X, σ_X^2, μ_Y, σ_Y^2 and ρ. Then, given $X = x$, Y is normally distributed with

$$E[Y \mid X = x] = \mu_Y + \rho\sigma_Y \left(\frac{x - \mu_X}{\sigma_X} \right),$$
$$V[Y \mid X = x] = (1 - \rho^2)\sigma_Y^2.$$

Theorem: If X and Y are bivariate normal and uncorrelated, then they are independent.

KNOX COLLEGE

The Multivariate Normal Distribution

But we can take this further!

The random variables $\boldsymbol{X} = [X_1, X_2, \ldots, X_n]^T$ have a **multivariate normal distribution** if their joint PDF is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left\{ -\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \right\},$$

where $\boldsymbol{\mu} = [\mu_1, \mu_2, \ldots, \mu_n]^T$ is the vector of means, Σ is the (invertible) covariance matrix, and $|\Sigma|$ is its determinant.

KNOX COLLEGE

The Multivariate Normal Distribution

The marginal distributions of a multivariate normal distribution are normal.

The conditional distributions of a multivariate normal distribution are normal.

Linear regression is closely tied to the bivariate normal: if X and Y are jointly normal, then $E[Y \mid X = x]$ is a straight line in x, and the variation of Y around that line is Normal with constant variance $(1 - \rho^2)\sigma_Y^2$.

KNOX COLLEGE